

WEEK 3

Marketing & Media Mix Modeling and ROI Optimization

From \$78M of spend to a reallocation you can defend in front of a CFO — with the uncertainty on the slide.

Andi Cupallari · Applied Data Science and AI in Pharma Commercial · Fall 2026 · Monday, September 28

Where we're going tonight

2.5 hours · one live notebook in two halves · one in-class exercise · Homework 1 released

0

Recap, mentors, and the Brand Lead's question

Which \$10M would you cut? · three families of attribution — 15 min

1

The MMM equation

Adstock, Hill saturation, the response curve; average vs. marginal — 25 min

2

Fitting an OLS MMM — live, part 1

Controls (ramp, season, competitor, access) · grid search · diagnostics — 35 min

3

Decomposition, ROI and uncertainty — live, part 2

Base vs. incremental · ROI table · bootstrap and grid sensitivity · grading against the truth — 30 min

4

Calibration and Bayesian MMM

A DTC-TV geo holdout · priors · PyMC-Marketing in 12 lines — 15 min

5

Pitfalls and pharma realities

Five ways MMM lies · six pharma quirks · when MMM is the wrong tool — 10 min

6

In-class exercise, Homework 1, next week

Adstock by hand · reallocation by hand · pick the tool — 20 min

00

Recap and the Brand Lead's question

Two weeks of 'what the manufacturer can see'. Tonight: what did its own spend do?

Three things from Weeks 1–2 that tonight depends on

If any of these is fuzzy, say so now — the model inherits all three

1

The payer decides whether a script fills

- ▶ Week 1: chooser ≠ payer ≠ consumer. Formulary status moves TRx with zero spend change.
- ▶ Tonight it is a control variable — and the one most MMMs leave out.

2

TRx is projected, not counted

- ▶ Week 2: the response variable is a stratified projection from a pharmacy panel, restated as the panel changes.
- ▶ Noise in y is fine for OLS; a restatement mid-series is not.

3

Contribution per TRx ≈ \$250

- ▶ Week 1's GLUCOREN P&L: net \$274 per TRx, contribution \$252.
- ▶ Tonight's ROI uses a \$250 placeholder. Finance owns this number; never invent it in a real engagement.

Mentor check: every team should have met its mentor by tonight. Problem statement + data plan due Monday, October 5 (one page, Week 2 template).

The Brand Lead's question — asked at every quarterly business review

GLUCOREN, three years after launch

**“We have spent \$78M promoting GLUCOREN since launch — about \$26M a year across reps, conferences, digital, email and TV.
Which \$10M would you cut? And how sure are you?”**

What we owe back

- ▶ A decomposition: how much TRx the brand generated on its own vs. from each channel
- ▶ ROI per channel — average (what we got) and marginal (what the next dollar gets)
- ▶ A reallocation with a stated risk posture, and the range around every number
- ▶ The list of what the model is not capturing, and the experiment that would settle it

What we will not pretend

- ▶ That a regression on 156 weekly rows is a causal experiment
- ▶ That an ROI to two decimal places means anything
- ▶ That the model can forecast a formulary change it has never seen
- ▶ That the highest-ROI channel is the one to grow (it usually is not)

Three ways to attribute sales to marketing — pick on purpose

Tonight is MMM. The right answer in industry is 'all three, calibrating each other'.

	Marketing Mix Modeling (MMM)	Multi-touch attribution (MTA)	Incrementality experiments
What it is	Top-down regression of weekly (or geo-weekly) outcomes on transformed spend + controls	Person-level, journey-based credit for digital touchpoints	Geo holdouts, matched DMAs, switchbacks, conversion-lift tests
Sees	Every channel incl. offline: reps, conferences, TV, samples; the system (base, season, competitor)	Only what is tracked: web, email, banners, search — one identity graph	One channel (or one change) at a time, cleanly
Blind to	Day-level tactics; person-level; anything that never varied in the data	Offline entirely; increasingly broken by privacy and cookie deprecation; no HCP-level consent in pharma	Everything not tested; slow; costs real revenue in dark cells
Data needed	2+ years weekly; spend by channel; controls (access, competitor)	Person-level event logs with a stable ID	Randomizable units (DMAs, territories, HCP lists) and patience
Pharma fit	The standard tool for annual planning and the QBR	Limited: HCP digital only; patient-level MTA is off-limits	Field-force experiments are old (territory holdouts); DTC geo tests are common

01

The MMM equation

Two transforms — carryover and saturation — turn spend into something a regression can see.

What we are actually fitting

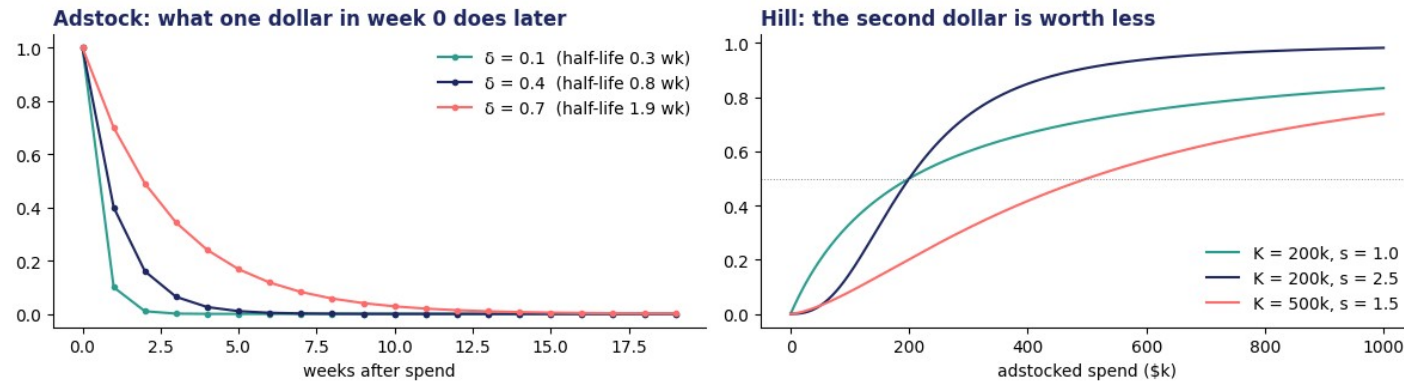
Read it left to right; every term is a decision you will have to defend

$$\text{TRx}_t = \beta_0 + \sum_c \beta_c \cdot \text{Hill}(\text{Adstock}(x_{c,t}; \delta_c); K_c, s_c) + \gamma \cdot \text{ramp}_t + \eta_1 \sin_t + \eta_2 \cos_t + \lambda \cdot \text{competitor}_t + \varphi \cdot \text{access}_t + \varepsilon_t$$

Term	Meaning	Estimated how	If you leave it out
$\beta_0 + \gamma \cdot \text{ramp}$	The base: what the brand does on its own (launch ramp, awareness, refills accumulating)	OLS (ramp midpoint and scale on a grid)	Channels that grew with the brand steal the ramp
$\text{Adstock}(x; \delta)$	Carryover: this week's spend works for weeks. δ = share carried to next week	Grid or prior — non-linear	Everything looks like same-week response; ROI understated for slow channels
$\text{Hill}(a; K, s)$	Saturation: K = half-saturation point, s = shape (S-curve if $s > 1$)	Grid or prior — non-linear	Linear response → marginal = average → the optimizer pours money into one channel
β_c	Maximum weekly TRx the channel can deliver (Hill tops out at 1)	OLS, given the transforms	—
$\text{season}, \text{competitor}, \text{access}$	What moves TRx and is not promotion	OLS	A channel absorbs the effect (tonight: the formulary win)
ε	Everything else. Structure in here = something missing	Residual diagnostics	You ship a wrong model

Adstock — spend echoes through time

Geometric form: $a_t = x_t + \delta \cdot a_{t-1}$. Half-life = $\ln 0.5 / \ln \delta$.



Typical δ by channel (weekly data)

- ▶ **0.6–0.8:** TV, sponsorships, KOL programs — slow build, long tail
- ▶ **0.3–0.6:** HCP detailing, conferences — a rep visit works for a month or two
- ▶ **0.0–0.25:** email, paid search, banners — gone by next week

Two variants you will meet

- ▶ **Weibull adstock:** lets the peak land after week 0 (a detail → next eligible patient → script → fill lag of 4–8 weeks)
- ▶ **$I_{\max}$:** PyMC-Marketing truncates the tail at $I_{\max}$ weeks; 8–12 is typical

Hill saturation — the second dollar is worth less than the first

$h(a) = a^s / (a^s + K^s)$. K is where the channel is at half its maximum; $s > 1$ gives a threshold before it works.

K — half-saturation point

- ▶ Spend (adstocked) at which the channel delivers 50% of its maximum
- ▶ Small K relative to spend = the channel is saturated: more money does little
- ▶ Tonight K is searched as a multiple of the channel's median adstocked spend

s — shape

- ▶ $s = 1$: concave from the first dollar (diminishing returns immediately)
- ▶ $s > 1$: S-shaped — a frequency floor before the channel converts (one TV impression does nothing; ten might)
- ▶ s also decides how sharp the marginal ROI falls beyond K

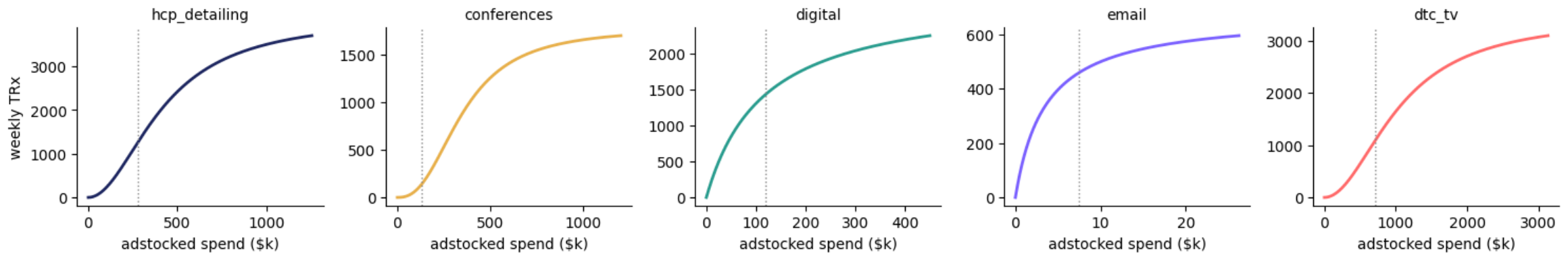
The response curve — adstock decides when, saturation decides how much

- ▶ **Average ROI** = total response ÷ total spend — the area under the curve so far. A reporting number.
- ▶ **Marginal ROI** = the slope of the curve at current spend — what the next dollar returns. A decision number.
- ▶ **Saturation guarantees they rank channels differently.** A saturated channel can have the best average and the worst marginal.

The one distinction that decides whether your recommendation is right

Tonight's fitted curves — dotted line = mean weekly adstocked spend when the channel is active

Fitted response curves — dotted line = mean weekly adstocked spend when active. Flat to the right of the line = saturated.



Email sits far to the right of its K: the best average ROI in the brand and almost nothing left to gain. Digital and detailing sit on the steep part. DTC TV is on the flat top of an S-curve with the biggest budget.

Reporting → average. Decisions → marginal. Confusing them in front of leadership produces confidently wrong reallocations — and it is the most common MMM error we see in QBR decks.

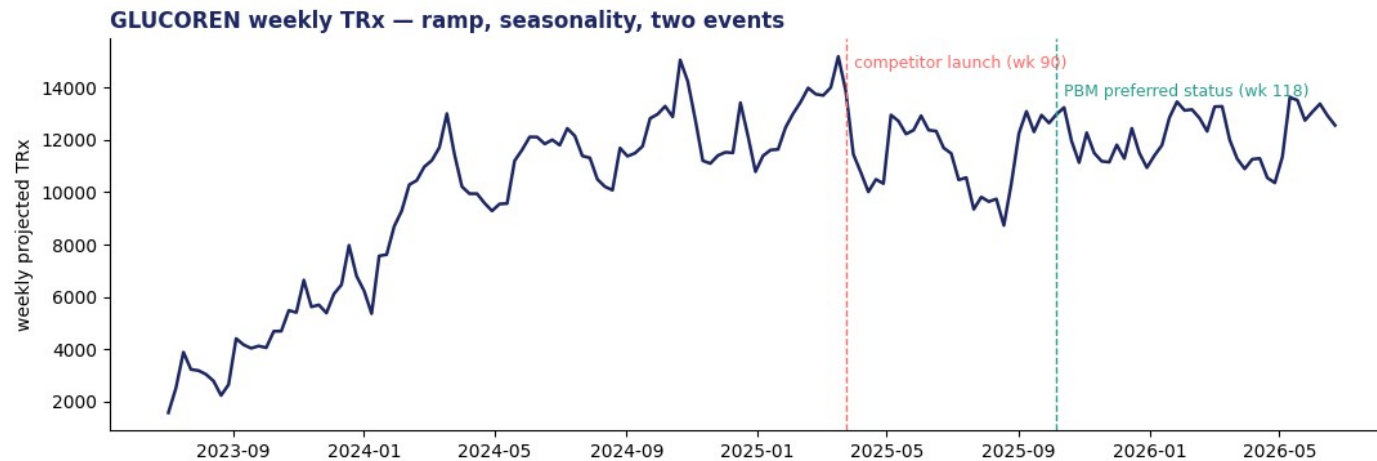
02

Fitting an OLS MMM — live

Part 1 of the notebook: controls, the grid, diagnostics. The fastest defensible model — if you respect the diagnostics.

GLUCOREN: 156 weeks, five channels, two events, one placeholder

glucoren_weekly_mmm.csv — projected national TRx and weekly spend; synthetic, from a known response model



\$78M

spend over the launch period

\$26M / yr

35% TV • 34% detailing • 18%
digital • 12% conferences • 1%
email

1.62M

TRx over 156 weeks

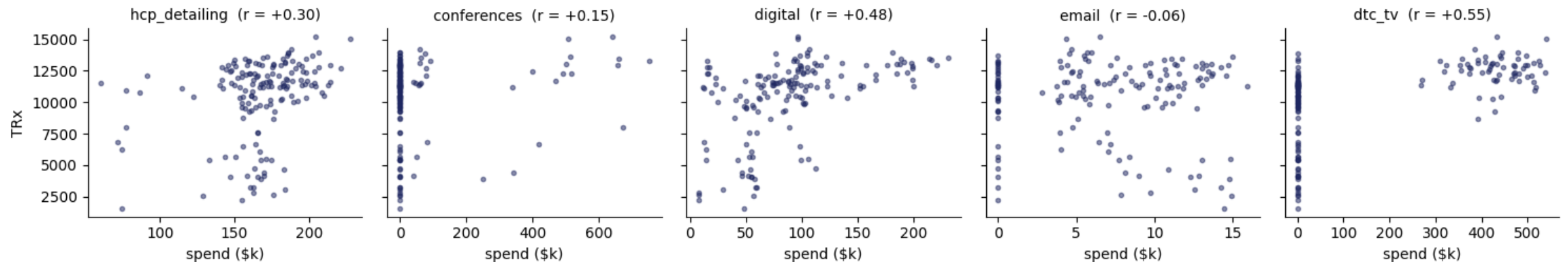
Column	What it is
trx	Weekly projected national TRx (Week 2's panel projection)
*_spend (5)	Weekly \$ by channel: detailing, conferences, digital, email, DTC TV
hcp_calls	Rep calls delivered — an alternative 'dose' for detailing
competitor_active	0/1 from week 90 (a competitor launched)
pbm_preferred_share	Share of commercial lives with preferred status; steps 0.38 → 0.61 at week 118

Detailing is steady, conferences are spikes, TV runs in 8-week flights, digital pulses and co-moves with TV, email is tiny. Each pattern is a different identification problem.

Same-week spend vs. TRx: weak, and for detailing negative

Marketing does not 'not work'. The plot is wrong: carryover, saturation and the launch ramp are all missing.

Same-week spend vs. TRx — weak, and in places negative. Marketing does not 'not work'; the plot is wrong.



Detailing was trimmed in year 3 while TRx kept rising on its base and on the formulary win — so a same-week scatter says reps hurt sales. Every brand team has seen a slide like this presented as evidence.

The fix is not a smarter scatter. It is the model on slide 8: transform the spend, control for what is not promotion, then read the coefficients.

In pharma, the controls are the model

Everything that moves TRx and is not promotion must be in the regression — or a channel will take the credit

Control	Tonight's feature	Why it is there	What happens if you leave it out (from the notebook)
Launch ramp	Logistic in t; midpoint and scale searched by AIC (media-blind fit is biased upward by early marketing)	New brands grow on their own: awareness, formulary wins, refills accumulating	Channels that grew with the brand (digital) become the ramp
Seasonality	Annual sine + cosine	Deductible resets, refill timing, flu season in some categories	A channel that spends in Q1 gets the January bump
Competitor launch	0/1 from week 90	Share loss that has nothing to do with our spend	The channels cut after the launch look ineffective
Formulary access	Share of lives with preferred status (0.38 → 0.61 at week 118)	Week 1: the payer decides whether the script fills	Competitor coefficient halves; detailing loses 18% of its credit; AIC worse by 79
Not modeled tonight	Copay program, samples, price changes, MLR content cycles, panel restatements	Each is a real driver; name them in the memo	They land in the residual — inspect it

The OLS recipe — six steps, in order

The order matters: controls before channels, diagnostics before decomposition, decomposition before ROI

1 Engineer the controls

Ramp, season, competitor, access. Fit the ramp media-blind first, then let AIC refine it.

2 Bound the grid with knowledge

δ ranges per channel (TV 0.4–0.7, email ≤ 0.25) — the OLS analogue of a prior. K as a multiple of median adstocked spend; $s \in \{1 \dots 2.5\}$.

3 Search, one channel at a time

Coordinate descent on AIC, two or three sweeps; ramp parameters in between. ~1,000 OLS fits, seconds.

4 Diagnose

Residuals (Durbin–Watson), VIF, signs, hold-out MAPE. Fix the data or the controls; never keep a negative media coefficient.

5 Decompose

$\beta_c \times \text{Hill}(\text{Adstock})$ per week, stacked on the base. This is the chart leadership wants.

6 ROI with ranges

Average and marginal per channel; block-bootstrap the coefficients; re-run the runner-up grid cells; then reallocate.

Where δ , K and s come from — grid search now, priors later

They are non-linear in the loss; OLS cannot solve them. Two honest ways to get them.

Grid search (tonight)

- ▶ **How:** for each channel, try every (δ , K, s) on a coarse grid with the other channels fixed; keep the AIC-minimizing cell; sweep again.
- ▶ **Pros:** fast, transparent, easy to explain to a brand team; no sampler to debug.
- ▶ **Cons:** no uncertainty on the winner; neighbouring cells within 2 AIC are indistinguishable and can imply different ROI (notebook §5c: detailing δ 0.40 vs 0.25, ROI 1.88 vs 1.76).
- ▶ **Rule:** bound the grid with channel knowledge; report the runner-up cells.

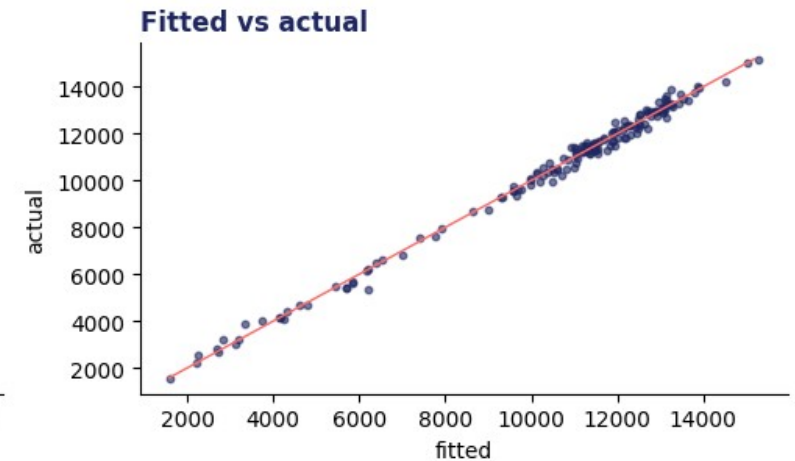
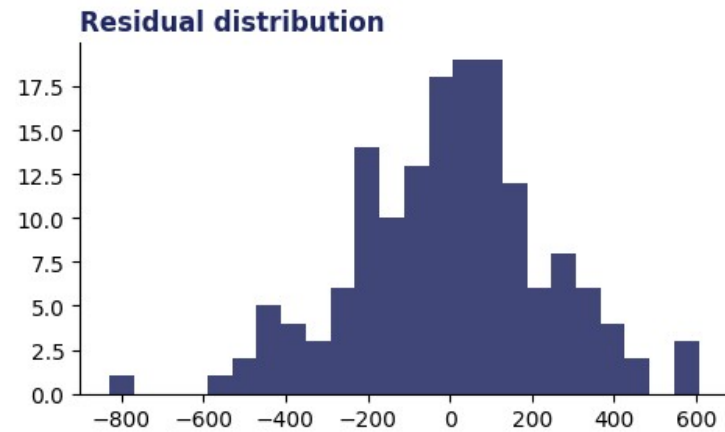
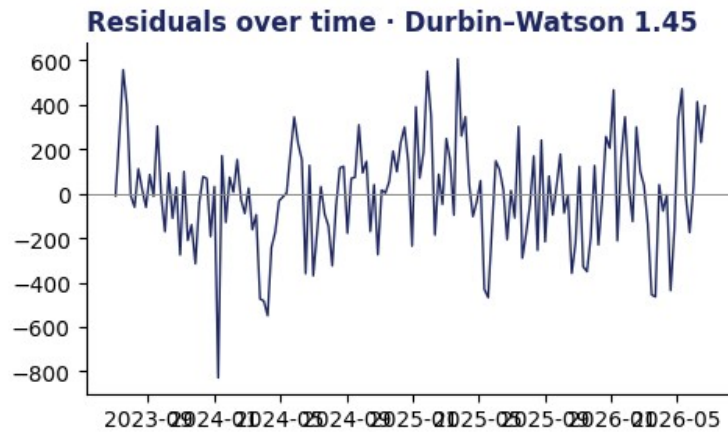
Bayesian priors (section 4)

- ▶ **How:** Beta(2,2) on δ , log-normal on K, gamma on s, half-normal on β ; MCMC gives posteriors on everything, including ROI.
- ▶ **Pros:** credible intervals for free; priors can encode a geo-lift result or last year's model; answers 'P(ROI_A > ROI_B)' directly.
- ▶ **Cons:** minutes of sampling, harder to debug, and priors you must be able to defend in a room.
- ▶ **Practice:** OLS for v1 ('is there a signal?'), Bayesian for v2 ('how sure, and what should we do?').

Tonight's winners: detailing δ 0.40 · K 1.5× · s 2.0 | conferences 0.55 · 8× · 2.5 | digital 0.25 · 1× · 1.0 | email 0.10 · 0.5× · 1.0 | DTC TV 0.70 · 1.5× · 2.0. The generating model used δ = 0.55 / 0.35 / 0.25 / 0.10 / 0.65.

Diagnostics — what stops you from shipping a wrong model

From the notebook: R^2 0.994 • Durbin-Watson 1.45 (mild positive autocorrelation — the ramped competitor effect is modeled as a step) • every VIF < 2.6 • hold-out MAPE 4.9%



Residuals

► White noise around zero. A slow wave = a missing control; autocorrelation (DW far from 2) = a missing trend or wrong δ .

VIF

► < 5 fine • 5–10 suspicious • > 10 not separately identified. Co-spend (digital with TV) and ramp-like channels are the usual causes.

Signs

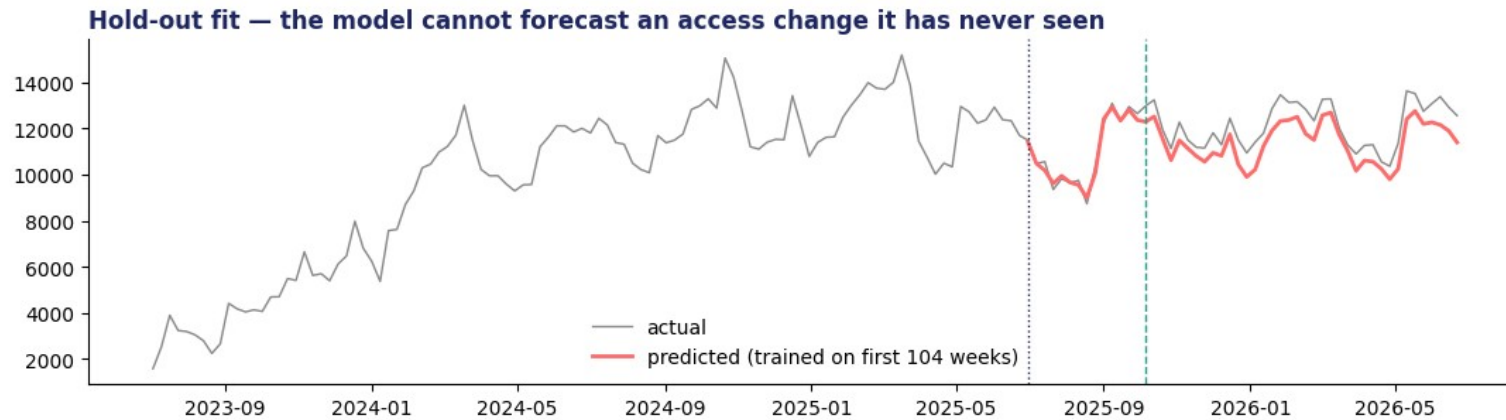
► Every media β positive. A negative detailing β is a confounder, not a finding — fix the controls, do not report it.

Hold-out

► Train on 104 weeks, predict 52: MAPE 1.6% before the formulary step, 6.1% after. The model cannot forecast an event it never saw.

The hold-out shows what an MMM is for — and what it is not for

Coefficients trained on weeks 1–104; the formulary win lands at week 118, inside the test window



Two caveats for the homework

- ▶ **1.** This hold-out tests the coefficients, not the grid: δ , K , s were chosen on all 156 weeks. The strict version re-runs the grid on the training window.
- ▶ **2.** An MMM explains the past. It does not forecast an access change, a recall or a label event it has never seen. Forecasting is Week 6; interrupted time series is the tool for one-off events.
- ▶ **Rule:** state both in the memo. A reviewer who finds them first stops reading.

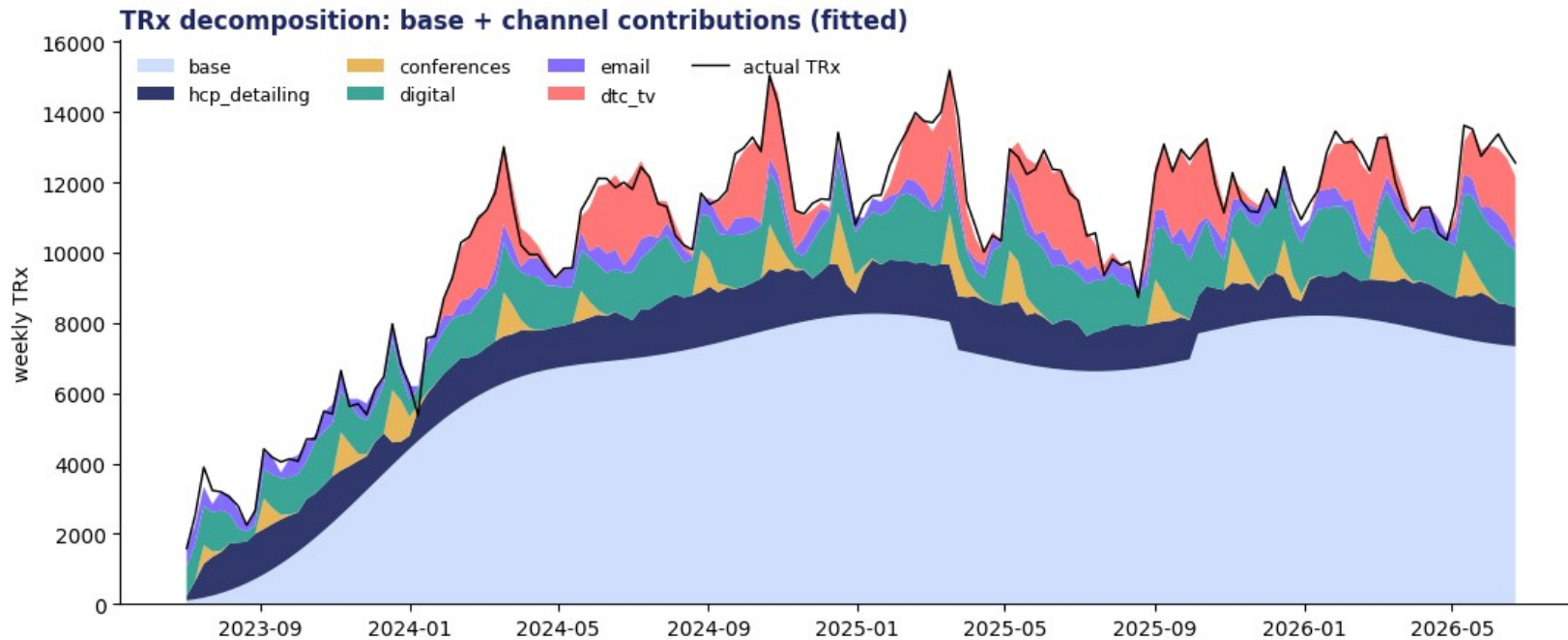
03

Decomposition, ROI and uncertainty — live

Part 2 of the notebook: the chart leadership wants, the table they will argue about, and the ranges that keep you honest.

Base vs. incremental — show this before any ROI number

Of 1.62M TRx: 61% base • 12% detailing • 13% digital • 8% DTC TV • 4% email • 2% conferences



How to read it

- ▶ Base = intercept + ramp + season + competitor + access. It is the brand's own momentum — and the payer's decisions.
- ▶ Each band is $\beta_c \times \text{Hill}(\text{Adstock})$ for that week. TV shows up as flights; conferences as spikes that decay.
- ▶ Black line = actual. Gap = residual. Here the fit is tight; in real data expect 3–6% unexplained.
- ▶ The generating model's base share was 60.7%. We recovered 60.8%.

ROI per channel — average vs. marginal, and the truth we happen to know

Contribution at \$250 per TRx. Marginal = +5% spend in every week of that channel, model re-evaluated.

Channel	Spend \$M	Incremental TRx	Avg ROI (fitted)	Avg ROI (truth)	Marginal ROI (fitted)	Marginal (truth)	Rank avg → marginal
Email	1.0	61k	14.7	11.1	4.6	3.7	1 → 1
Digital	14.1	208k	3.70	3.58	1.76	1.85	2 → 3
HCP detailing	26.1	196k	1.88	2.09	2.55	1.61	3 → 2
DTC TV	27.6	133k	1.21	1.28	1.18	1.04	4 → 5
Conferences	9.1	40k	1.09	1.00	1.19	0.80	5 → 4

What this brand teaches

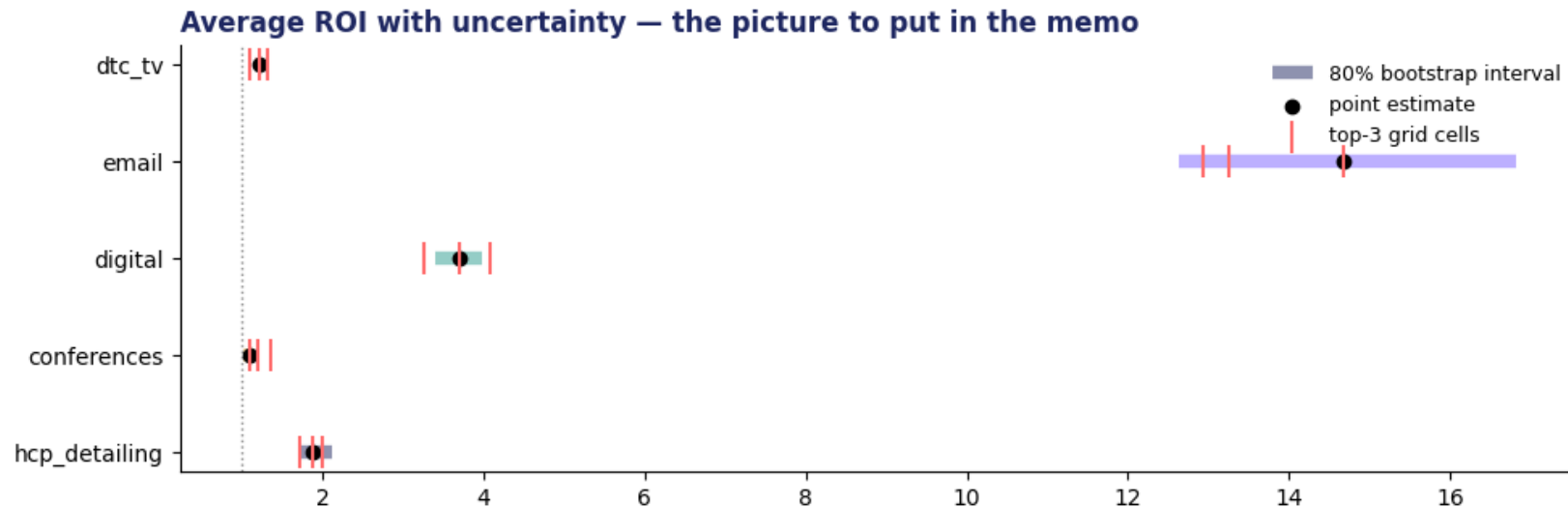
- ▶ Email: best ROI, \$1M budget, saturated — the answer to 'so put \$5M into email?' is an ROI of 1.4 on that \$5M (notebook \$11)
- ▶ TV: 35% of the budget at the lowest marginal return — cut, do not kill (the geo test says it works)

Where the method missed

- ▶ Detailing's marginal is over-stated (fitted $s = 2.0$ vs. true 1.3): steady spend gives the curve little to learn from
- ▶ Conferences: episodic spend — compute ROI by event, not weekly average

Ranges, not points — the picture that belongs in the memo

80% moving-block bootstrap intervals (300 refits, 8-week blocks) plus the ROI implied by the three best grid cells per channel



Two sources, two tools

- ▶ **Sampling error:** bootstrap the residuals in blocks (keeps autocorrelation), refit, read percentiles. Narrow here because the fit is tight.
- ▶ **Specification error:** neighbouring grid cells ($\Delta AIC < 2$) imply different ROI — the coral ticks. Wider than the bootstrap for detailing.
- ▶ **Neither** captures a missing control. Only a design change — or an experiment — does that.

Report 'digital 3.4–4.0' rather than '3.70'. It lands better with a CFO, and it is more honest.

Grading against the truth: the ranking is right, the levels are $\pm 10\text{--}30\%$, the decay is within a grid step

Synthetic data lets us do what a real engagement never can. The point is to see how the method misses.

Channel	Fitted TRx	True TRx	Error	Fitted δ / s	True δ / s	Why
HCP detailing	196k	218k	-10%	0.40 / 2.0	0.55 / 1.3	Steady spend: the curve's shape is barely identified; the S-curve over-states the marginal
Conferences	40k	36k	+9%	0.55 / 2.5	0.35 / 1.8	Spikes identify β well; the tail is confounded with the following weeks' detailing
Digital	208k	202k	+3%	0.25 / 1.0	0.25 / 1.0	Pulses and dark windows give the grid what it needs
Email	61k	46k	+33%	0.10 / 1.0	0.10 / 1.5	Tiny channel; wide interval; a saturated channel looks like the intercept
DTC TV	133k	141k	-6%	0.70 / 2.0	0.65 / 1.6	Flights on/off are the ideal identification pattern

The pattern generalises: channels that vary (flights, pulses) are measurable; channels that never vary (an always-on field force) are not — however big. Variation in spend is the price of measurement. That is the argument for planned experiments.

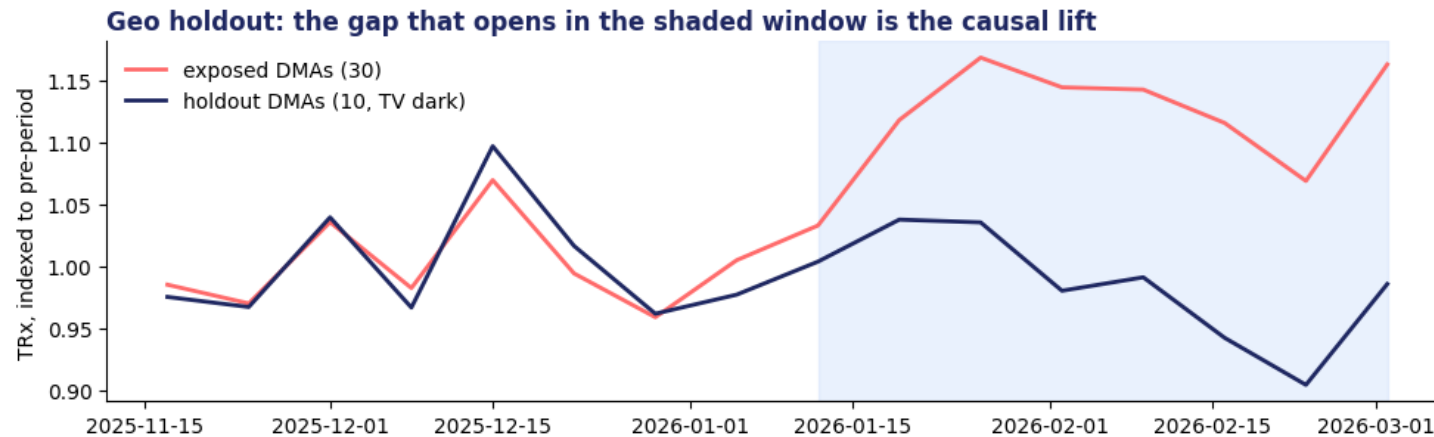
04

Calibration and Bayesian MMM

Models without calibration are confident lies. One clean experiment per major channel per year beats a fancier model.

The DTC-TV geo holdout: flight 7 dark in 10 of 40 DMAs

Difference-in-differences on the DMA panel: 8 pre-weeks, 8 test weeks. glucoren_geo_lift_dtc_tv.csv



	Pre (8 wk)	Test (8 wk)	Growth
Exposed (30 DMAs)	74,139	82,959	1.119
Holdout (10 DMAs)	17,719	17,453	0.985

Result

- ▶ Lift = $82,959 - 74,139 \times 0.985 = 9,933$ TRx (13.6%) in exposed DMAs
- ▶ TV spend in those DMAs: \$2.36M → experiment ROI 1.05
- ▶ MMM's view of the same flight and DMAs: 0.97. Truth: 1.08.
- ▶ Agreement inside the interval → the TV coefficient is credible. Disagreement → the experiment wins; it has the cleaner identification.

What could make the difference-in-differences wrong

- ▶ Spillover: TV signal leaks across DMA borders (holdout is partly exposed → lift understated)
- ▶ Non-parallel trends: the ten dark DMAs were already growing differently (check the pre-period)
- ▶ Too few units: 10 dark DMAs is thin — real tests use matched pairs and a power calculation before the flight

Calibration — the triangle every MMM needs

Each corner fixes a weakness of the other two. Build a portfolio of past lifts; each one tightens the next model.

MMM

System view, every channel, annual planning. Weak on causality.

Experiments

Causal, one channel at a time. Expensive in dark cells and time.

Digital attribution / CRM

Fast, tactical, person-level for HCP digital. Blind offline; privacy-limited.

How the experiment enters the model

- ▶ OLS: check the coefficient against the lift; if outside the interval, constrain the grid or add the geo-week rows to the regression (a panel MMM)
- ▶ Bayesian: turn the lift into an informative prior on β_{TV} (mean and sd from the test); the posterior then blends data and experiment
- ▶ Either way: document which experiment, which flight, which DMAs

Cadence and cost

- ▶ One geo-lift per major channel per year is a reasonable target; a \$2–3M flight with a 25% dark cell costs ~\$0.6M of TV and some TRx
- ▶ Field-force tests are older than MMM: territory-level detailing holdouts, sample stops
- ▶ Digital lift tests (conversion lift, PSA holdouts) are cheap — run them continuously

Bayesian MMM — why the slowdown is worth it, and PyMC-Marketing in 12 lines

Same model class. Priors replace the grid; the sampler returns posteriors on δ , K , s , β — and therefore on ROI.

Three reasons

- ▶ **Identifiability made explicit:** δ and β trade off (a longer tail with a smaller β fits the same curve). OLS picks one pair and pretends; the posterior shows the ridge.
- ▶ **Evidence outside the data:** last year's geo-lift, ten years of detailing literature, the agency's prior model — all become priors instead of being ignored.
- ▶ **Decision-native questions:** 'P(ROI_digital > ROI_TV + 1)' or 'probability the reallocation loses money' come straight from the draws.

```
from pymc_marketing.mmm import MMM, GeometricAdstock, LogisticSaturation

mmm = MMM(
    date_column="week",
    channel_columns=[f"{c}_spend" for c in CHANNELS],
    control_columns=["ramp", "season_sin", "season_cos",
                    "competitor", "pbm_preferred"],
    adstock=GeometricAdstock(l_max=10),
    saturation=LogisticSaturation()
)
mmm.fit(X=X, y=y, chains=2, draws=1000, tune=1000,
        target_accept=0.9, random_seed=7)
mmm.plot_posterior_predictive(); mmm.plot_channel_contributions_grid()
```

Notebook §12 has this cell behind a BAYES flag; sampling takes minutes. Homework 1's bonus asks for 80% credible intervals on ROI and one posterior-probability statement. Syllabus reading: Jin et al. (Google, 2017) is the paper behind the carryover and shape priors.

05

Reallocation — the answer to the question

Equalize marginal ROI under constraints, in small steps, with a stopping rule. Then say it like a strategist.

Equalize marginal ROI — with floors, caps and a stopping rule

Notebook §11: a greedy optimizer that moves \$0.25M at a time from the lowest to the highest marginal ROI and stops when the gap is inside the noise

The textbook rule

- ▶ At the optimum every channel has the same marginal ROI
- ▶ Move budget from low-marginal to high-marginal channels in small steps and re-evaluate — the curves move under your feet
- ▶ You never reach the optimum; you stop when the gap is smaller than your uncertainty (tonight: 0.15)

Tonight's constraints

- ▶ Floors: detailing $\geq 85\%$ (a field force is not a dial); conferences $\geq 60\%$ (contracted sponsorships)
- ▶ Caps: email $\leq 2.5\times$ (list size, fatigue, CAN-SPAM); digital $\leq 1.8\times$ (inventory, MLR-approved content); detailing $\leq 1.3\times$ (hiring)
- ▶ Total budget fixed; the CFO's question was 'which \$10M', not 'how much more'

Channel	Current \$M	Conservative ($\leq$ \$5M moved)	Aggressive ($\leq$ \$15M moved)
HCP detailing	26.1	+4.25 $\rightarrow$ 30.35	+7.75 $\rightarrow$ 33.85 (cap)
Conferences	9.1	-1.50 $\rightarrow$ 7.57	-3.25 $\rightarrow$ 5.82 (floor)
Digital	14.1	— $\rightarrow$ 14.08	+1.75 $\rightarrow$ 15.83
Email	1.0	+0.75 $\rightarrow$ 1.79	+1.50 $\rightarrow$ 2.54 (cap)
DTC TV	27.6	-3.50 $\rightarrow$ 24.06	-7.75 $\rightarrow$ 19.81
Result (3-year total)		+27k TRx $\approx$ +\$6.8M contribution, +1.7%	+44k TRx $\approx$ +\$11.0M contribution, +2.7% (stopped at \$11M)

So — which \$10M would you cut?

The recommendation, stated the way it should be stated

“Cut \$7–8M of DTC TV (drop two of the eight flights) and \$2–3M of conferences; move \$4–8M to the field and \$1M to email. Expected +1.7 to +2.7% TRx on a flat budget. Our TV number is validated by the Q4 geo test; our detailing number is the least certain and is the one to test next.”

1 Say the decomposition first

- ▶ Base, channels, competitor, access. They have to see the system before they hear a ratio.
- ▶ Then the ROI table — with ranges.

2 Two scenarios, one risk posture each

- ▶ Conservative and aggressive; let the room choose.
- ▶ List what the model is not capturing: copay program, samples, MLR timing, panel restatements.

3 End with the experiment

- ▶ 'A 20-territory detailing holdout in Q1 would resolve the field number for ~\$0.4M of lost calls.'
- ▶ Name the cost and the decision it unlocks.

06

Pitfalls and pharma realities

Five ways MMM lies to you, six things that wreck a generic MMM in pharma, and when it is the wrong tool.

Five ways MMM lies to you — and the fix for each

Every one of these appeared in tonight's notebook on purpose

Lie	What it looks like	Tonight	Fix
Identifiability	Two (δ, β) pairs fit identically; ROI differs	Detailing δ 0.40 vs 0.25: Δ AIC 1.6, ROI 1.88 vs 1.76	Bound the grid with knowledge; priors; report runner-up cells; experiment
Multicollinearity	Channels co-spend; VIF > 10; signs flip	Digital co-moves with TV ($r = 0.46$); free-grid runs push VIF > 20	Ridge, merge channels, or vary the spend on purpose
Seasonality vs. media	Q1 spend gets credit for the January bump	Season terms in before channels	Explicit seasonality first; check residuals by month
Missing control	A channel absorbs an event	Formulary win: competitor coefficient halves without it	Access, competitor, price, copay in the model; residual plot
Selection bias	Spend rises when growth is already strong	Detailing cut after the competitor launch (managers chase momentum)	Instrument, DAG-aware controls, or flag it — and run an experiment

Six pharma-specific things that wreck a generic MMM

A consumer-goods MMM template will get every one of these wrong

Long latency

Detail → physician thinks → next eligible patient → script → fill → claim → panel projection. 4–8 weeks. Weibull adstock or a lagged term; weekly data is the minimum.

Formulary and prior auth

A tier change moves TRx overnight with zero spend change. Access covariates or event dummies — from the payer file, not the agency.

Copay programs and samples

Drive volume but are not 'media'. Separate levers with their own transforms — and gross-to-net implications (Week 1).

Detailing is not a dollar

It is calls × content × coverage × reach. Use call equivalents or reach as the dose; test hcp_calls vs. spend (notebook 'try before Week 4').

Specialty brands are small

A few thousand prescribers: the law of large numbers does not apply. Hierarchical models by territory; HCP-level models (Weeks 4–5).

MLR and compliance lag

DTC content takes 6–10 weeks of review. 'Reallocate to TV by next quarter' may not be feasible; say so in the memo.

When MMM is the wrong tool

Knowing this is worth more than knowing PyMC

Situation	Why MMM fails	Use instead	Course week
Brand-new launch, < 1 year of data	No cycles to identify channels; the ramp swamps everything	Analog launches + Bayesian priors; launch-archetype benchmarks (Week 1's IQVIA framework)	1, 6
Day-level digital decisions	Weekly cadence too slow; channels too granular	Digital attribution, contextual bandits	5
HCP-level next best action	MMM is brand-level; it cannot say who to call	Uplift models, affinity scores, orchestration	4–5
One-off causal event (label change, recall, FDA action)	Treated as noise or absorbed by a channel	Interrupted time series, synthetic control	6
Deciding total budget, not mix	MMM allocates a given budget; the curve's level is the least certain thing it estimates	Financial model + experiments on the margin	—

What the industry says it wants from this — and why it still struggles

From the IQVIA papers in the reading pack

“Life sciences organizations still need to prepare more effectively for high-value customer interactions, allocate spend with greater precision, understand care pathways and patient journeys, and respond to access barriers and customer behavior. What has changed is the tolerance for delay and ambiguity. Senior decision makers need answers faster, with context they can trust and enough confidence to act.”

— IQVIA, “AI Alone Will Not Modernize the Life Sciences Enterprise” (Silvis, 2026)

58%

of commercial leaders report $\geq 2\times$ one-year ROI from AI initiatives (IQVIA 2025 survey, n = 107)

11%

say their data is 'fully sufficient' for commercial AI; 52% 'mostly'

\$53M

average first-year promotional spend for a top-25 launch vs. \$17M for emerging biopharma (IQVIA, 605 launches)

\$75k+

found by an IQVIA audit in one health system's paid search spent on irrelevant keywords and geographies in a year

The measurement question is the same at every scale: a \$25M brand budget, a \$2M AI program, or a hospital's paid-search account. 'What did the spend do?' is answered by a model with controls, a range, and an experiment — or it is not answered.

Exercise: adstock by hand, reallocate by hand, pick the tool (20 min, groups of 3–4)

Handout: Week_03_InClass_Exercise. One group presents each part in the debrief.

A Part A • Transforms by hand (6 min)

- ▶ Six weeks of spend for one channel; compute geometric adstock at $\delta = 0.5$, then Hill at $K = 300$, $s = 1$
- ▶ Then: which single week's spend would you move to raise the six-week total most, and why?

B Part B • Reallocate by hand (8 min)

- ▶ A response-curve table for three channels (TRx at 80 / 100 / 120% of current spend)
- ▶ Compute average and marginal ROI; move \$2M from the worst marginal to the best; recompute
- ▶ Then: the CFO asks for \$4M of savings instead

C Part C • Pick the tool (6 min)

- ▶ Six situations a GLUCOREN brand team faced this year
- ▶ For each: MMM, experiment, digital attribution, interrupted time series, or 'none — it is a data problem'; and the control you would need

Answer key in the instructor package; students receive the worked solution after the debrief.

Homework 1 — released tonight, due Monday, October 12, 11:59 PM ET

10% of the final grade • +10 bonus • same data as tonight • Homework_1_MMM.docx has the full brief and rubric

Deliverables (one zip)

- ▶ **Notebook:** EDA → adstock + Hill → controls → OLS with a grid → diagnostics incl. hold-out → decomposition → average and marginal ROI → bootstrap or grid-sensitivity ranges
- ▶ **Memo (2 pages, PDF):** to the GLUCOREN Brand Lead — TL;DR, decomposition chart, a reallocation with a risk posture, what the model misses, one experiment with a cost
- ▶ **Bonus (+10):** PyMC-Marketing fit; 80% credible intervals on ROI; one posterior-probability statement; a paragraph on whether the recommendation changes

Rubric (100 + 10)

- ▶ EDA + controls • 10
- ▶ Adstock + Hill • 15
- ▶ OLS MMM + diagnostics • 25
- ▶ Decomposition + ROI with ranges • 15
- ▶ Memo • 35
- ▶ Bayesian bonus • +10

Rules: numpy seed 7; the notebook runs top-to-bottom on a clean kernel; no code in the memo; AI use disclosed in a one-paragraph appendix (syllabus policy); late work not accepted. The instructor truth files are not distributed — do not go looking for them.

Before Week 4 (Oct 5): readings, data plan, homework started

Week 4: HCP and patient segmentation — decile vs. behavioral vs. needs-based; k-means, GMM, UMAP/HDBSCAN on claims features.

Do

- ▶ **Read:** Week 3 reading pack (~35 min) and the MMM Field Guide (reference — keep it next to the homework).
- ▶ **Read (syllabus):** Jin et al., “Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects” (Google, 2017) — §§1–3; skim the PyMC-Marketing MMM example notebook.
- ▶ **Watch / read (optional, PMSA):** Axtia “Promotion Response Modeling for Pharma”; “Marketing-Mix Methodologies to Account for Managed Care Dynamics.”
- ▶ **Submit:** problem statement + data plan (one page) by Oct 5.
- ▶ **Start Homework 1** this week; the PyMC bonus needs sampling time.

Bring

- ▶ **One channel from your term project** (or from any brand you know) and the pattern of its spend: steady, bursty, flighted, growing? What does that mean for measuring it?
- ▶ **Your answer to:** “what would force us to retrain this model?” (a formulary change, a competitor, a panel restatement, a pandemic — the PMSA 2021 post-COVID recalibration sessions are the reference).
- ▶ **For Week 4:** scikit-learn, umap-learn, hdbscan installed.

Office hours Mondays 5:00–6:45 PM • ac5562@columbia.edu

Questions?

Five things to leave with:

1 · Adstock and saturation are not optional. 2 · In pharma the controls are the model — ramp, season, competitor, access.
3 · Average ROI reports; marginal ROI decides. 4 · Ranges, not points — and a channel that never varies cannot be measured. 5 · Calibrate:
where the experiment and the model disagree, the experiment wins.

Next week — HCP and patient segmentation · Homework 1 due October 12